

TEST 1

A READING

Task 1. Read the article. The summary of the article is provided after the text. Sentences of the summary are mixed up. For questions (1-10), put sentences (A-J) of the summary in the chronological order of events. There is an example at the beginning (0).

Example: |0|K|

0. *Computer animation can be created with a computer and animation software. Some impressive animations can be achieved even with basic programs; however, the rendering can take a lot of time on an ordinary home computer. Because of this, video game animators tend to use low resolution renders, such that the graphics can be rendered in real time on a home computer. Photorealistic animation would be impractical in this context.*

(1) Animation is the rapid display of images of 2D or 3D artwork or model positions in order to create an illusion of movement. It is an optical illusion of motion due to the phenomenon of persistence of vision. It involves the appearance of motion caused by displaying still images one after another. Often, animation is used for entertainment purposes.

(2) In addition to its use for entertainment, animation is considered a form of art. It is often displayed and celebrated in film festivals throughout the world. Also used for educational purposes, animation has a place in learning and instructional applications as well.

(3) Cartoon animation is often considered to be animation in its classic form. The animated cartoon made its debut in the early part of the 20th century and calls for the use of 24 different drawings per second. In traditional animated cartoons, frames are hand drawn.

(4) Computer animation or CGI animation is the process used for generating animated images by using computers. It is a subfield of animation and computer graphics. The more general term CGI (computer-generated imagery or computer-generated imaging) encompasses both static scenes and dynamic images, while computer animation only refers to moving images.

(5) Modern computer animation usually uses 3D computer graphics, although 2D computer graphics are still used for stylistic, low bandwidth, and faster real-time renderings. Sometimes the target of the animation is the computer itself, but sometimes the target is another medium, such as film.

(6) Computer animation is essentially a digital successor to the stop motion techniques used in traditional animation with 3D models and frame-by-frame animation of 2D illustrations. Computer generated animations are more controllable than other more physically based processes, such as constructing miniatures for effects shots or hiring extras for crowd scenes, and because it allows the creation of images that would not be feasible using any other technology. It can also allow a single graphic artist to produce such content without the use of actors, expensive set pieces, or props.

(7) To create the illusion of movement, an image is displayed on the computer monitor and repeatedly replaced by a new image that is similar to it, but advanced slightly in time (usually at a rate of 24 or 30 frames/second). This technique is identical to how the illusion of movement is achieved with television and motion pictures.

(8) For 3D animations, objects (models) are built on the computer monitor (modeled) and 3D figures are rigged with a virtual skeleton. For 2D figure animations, separate objects (illustrations) and separate transparent layers are used, with or without a virtual skeleton. Then, the limbs, eyes, mouth, clothes, etc. of the figure are moved by the animator on key frames. The differences in appearance between key frames are automatically calculated by the computer in a process known as tweening or morphing. Finally, the animation is rendered.

(9) For 3D animations, all frames must be rendered after modeling is complete. For 2D vector animations, the rendering process is the key frame illustration process, while tweened frames are rendered as needed. For pre-recorded presentations, the rendered frames are transferred to a different format or medium such as film or digital video. The frames may also be rendered in real time as they are presented to the end-user audience.

(10) Low bandwidth animations transmitted via the internet (e.g. 2D Flash, X3D) often use software on the end-users' computer to render in real time as an alternative to streaming or pre-loaded high bandwidth animations.

- A. 3D and 2D computer graphics are used for different purposes.
- B. The creation of illusion of movement is similar to the illusion achieved with television and motion pictures.
- C. Computer animation and computer generated imaging are different concepts.
- D. Frames may be rendered as film or in real time.
- E. Animation is the process of creating an illusion of movement.
- F. It is possible to create computer generated animations by a single specialist.
- G. A virtual skeleton is used for 3D animation.
- H. Animation implies multiple uses.
- I. Special software is needed to transmit low bandwidth animations.
- J. The classical form of animation is represented by cartoon animations.
- K. Animation software is used to create computer animation.

(each 2) ____ /

20

B USE of LANGUAGE

Task 2. Read the text and choose the most appropriate word from the list (A-L) for each gap (11-20). There are two extra words that you do not need to use. Use only one word in each gap.

Cybernetic Models' Application

Cybernetic models are employed in the specification of controllers and regulators for industrial plants, (11) _____, and so on. They are widely used in (12) _____ the proper relationship between a man and a machine, for example, in the design of vehicle-control systems. Another field of (13) _____ is teaching or training, which is interpreted as the control of a human learning process. A training instructor may be partially or wholly replaced by a suitable machine. In operational research, cybernetic models are used to specify stockholding (14) _____, process, and assembly programming. They are also used in a normative fashion; for example, the management of a business enterprise is often modelled as a game-like decision and control process. At a neuro physiological level, cybernetic models have been used to explain many aspects of the working of a (15) _____. Outside the brain, cybernetic principles are widely used to interpret the control of (16) _____ functions. Within psychology, it is possible to explain several classes of behaviour and (17) _____ in terms of hierarchies of control systems. At a (18) _____ level, cybernetic ideas are applied to interpersonal (19) _____ such as conversations, the communicative behaviour of small groups, etc. Ethnologists also use cybernetics freely, especially in dealing with population (20) _____ control systems and the regulation of reproduction.

A application
B density
C macroscopic
D schemes

E governance
F navigation
G cognition
H bodily

I determining
J robotic
K brain
L interactions

(each 1) __/10

Task 3. Read the text below. For the gaps (21-30) choose the most appropriate answer (A, B, C, or D).

When Machines Outsmart Humans

The distinction (21) _____ man and machine is rather blurred. The possibility that machines will (22) _____ the equivalent of human feelings and emotions is pure fantasy. And (23) _____, we cannot ignore advancing machine (24) _____ implications about the human mind. Think back to IBM's Deep Blue defeat of Garry Kasparov in 1997. One pivotal move that (25) _____ the match in favour of Deep Blue prompted Kasparov to accuse IBM of human intervention. He highlighted the essential cognitive dissonance that we will face (26) _____ machines get smarter. Kasparov couldn't believe that he (27) _____ by a computer because he felt the play, for him it was a sign of (28) _____ intelligence. But he was wrong — years later

it was revealed by Deep Blue's co-creator that the triumphant move had been a result of a (29) _____ bug. When presented with several options, Deep Blue could not make a definitive decision, so made a (30) _____ move that rattled Kasparov.

- | | | | | | | | | |
|----|---|-----------------|---|-------------|---|------------------|---|-----------------|
| 21 | A | Among | B | Between | C | Within | D | Inside |
| 22 | A | Overtake | B | Acquire | C | Imperil | D | Eclipse |
| 23 | A | Although | B | Despite | C | Yet | D | in spite |
| 24 | A | intelligence's | B | Intelligent | C | Intelligence | D | intelligences |
| 25 | A | Exchanged | B | Replaced | C | Mutated | D | Shifted |
| 26 | A | so as | B | as if | C | such as | D | As |
| 27 | A | has been beaten | B | was beaten | C | was being beaten | D | had been beaten |
| 28 | A | Elder | B | Superior | C | Crucial | D | Excellent |
| 29 | A | Software | B | web page | C | Design | D | Code |
| 30 | A | Planned | B | Definite | C | Random | D | Particular |
- (each 1) __/10

C WRITING

Task 4. Fill in the gaps (31-40) with the word which corresponds to the context of the text the most and makes its parts logical.

The Roots of Cybernetic Theory

The word *cybernetics* was first used in the context of “the (31) _____ of self-governance” by Plato in *The Laws* to signify the (32) _____ of people. The word “cybernétique” was also used in 1834 by the (33) _____ André-Marie Ampère to denote the sciences of government in his (34) _____ system of human (35) _____. The first artificial automatic regulatory (36) _____, a water clock, was invented by the mechanic Ktesibios. He and others such as Heron and Su Song are considered to be some of the first to study cybernetic (37) _____. Contemporary cybernetics began as an interdisciplinary study connecting the (38) _____ of control systems, electrical network theory, mechanical engineering, logic (39) _____, evolutionary biology and neuroscience in the 1940s. Electronic control systems originated with the 1927 work of Bell Telephone Laboratories engineer Harold S. Black on using negative (40) _____ to control amplifiers.

(each 1) __/10

Total	A+B+C
_____/50	